Free Energy Functionals in the Phase-Field Model

of Pure-Melt Solidification

Study notes on Bhagat et al., *Engineering with Computers* 39 (2023) 2345–2363, Sect. 2.2

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The question. Why are two different free energy functionals used to derive the phase-field equation and the temperature equation?
The answer in short. If we insisted on one functional for consistency, it would have to be the Lyapunov functional Π_L. That forces the coupling term in the ϕ equation to be tied to the latent-heat function $h(\phi)$. Using a separate *isothermal* free energy functional Π for the ϕ equation, and energy conservation for the temperature equation, lets the coupling and the latent heat be chosen independently. Both routes give the same sharp-interface Stefan problem; the two-functional route converges on coarser grids and is therefore much cheaper. The price is that no single functional is guaranteed to decrease.

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1 Notation: the six objects

The same letters mean different things in Bhagat et al. and in Karma & Rappel. These notes use the letters in the last column throughout.

Role	Bhagat et al.	Karma & Rappel	These notes
Double-well potential	$g(\phi)$	$f(\psi)$	g
Temperature tilt (coupling function)	$h(\phi)$	$g(\psi)$	p
Latent-heat (enthalpy) function	implicit in $U = u - \frac{1}{2}\phi$	$h(\psi)$	h
Bulk free energy density	$f(\phi, u)$	$f + \lambda u g$	f
Isothermal free energy functional	Π	F_{iso}	Π
Lyapunov functional	Π_L	F	Π_L

Here $\phi = +1$ is solid and $\phi = -1$ is liquid; $u = c_p(T - T_m)/L$ is the scaled temperature.

$g(\phi) = -\frac{1}{2}\phi^2 + \frac{1}{4}\phi^4$ **— double well.** Two equal valleys at $\phi = \pm 1$. It makes solid and liquid the only stable states. Balanced against the gradient term $\frac{1}{2}\lambda^2|\nabla\phi|^2$, it sets the interface width $\sim \lambda$ and the surface energy γ, the source of the Gibbs–Thomson effect.

$p(\phi) = \phi - \frac{2}{3}\phi^3 + \frac{1}{5}\phi^5$ **— temperature tilt.** Multiplied by ξu, it lowers one valley and raises the other: $u < 0$ favours solid, $u > 0$ favours liquid. It is the diffuse-interface version of the driving force $L\Delta T/T_m$. Since $p'(\phi) = (1 - \phi^2)^2$ vanishes at $\phi = \pm 1$, the valleys stay exactly at ± 1 for every temperature.

$h(\phi)$ **— latent-heat function.** Says how much latent heat a point holds, through $U = u - \frac{1}{2}h(\phi)$ (Section 3). Bhagat et al. use $h = \phi$, a linear interpolation of the latent heat across the interface.

$f(\phi, u) = g(\phi) + \xi u\, p(\phi)$ **— bulk free energy density.** The tilted double well: which phase is preferred at a point, at the local temperature.

Π **— isothermal free energy functional** (Bhagat Eq. 3a):

$$\Pi[\phi, u] = \int_\Omega \left[f(\phi, u) + \tfrac{1}{2}\lambda^2(\mathbf{n})|\nabla\phi|^2 \right] dV.$$

Π_L **— Lyapunov functional:**

$$\Pi_L[\phi, U] = \int_\Omega \left[g(\phi) + \xi u^2 + \tfrac{1}{2}\lambda^2(\mathbf{n})|\nabla\phi|^2 \right] dV, \qquad u = U + \tfrac{1}{2}h(\phi).$$

> h does not appear explicitly in Π_L. It is hidden inside u: the independent fields of Π_L are ϕ and the enthalpy U, and $u = U + \frac{1}{2}h(\phi)$ depends on both.

2 Building block: Gibbs free energy

Gibbs free energy is a *definition* that puts two competing tendencies into one number:

$$G = H - TS.$$

Low enthalpy H (tight bonds) favours the solid; high entropy S (disorder) favours the liquid, more so at high T. **At fixed temperature, the phase with lower G wins.** Comparing liquid and solid at the same T,

$$\Delta G = G_l - G_s = \Delta H - T\Delta S.$$

At T_m the phases coexist, $\Delta G = 0$, so $\Delta S = L/T_m$; below T_m the driving force for freezing is $\Delta G = L\,\Delta T/T_m$. The rule "lower G wins" holds only at constant temperature—the origin of the word *isothermal* below.

3 Derivation of $U = u - \frac{1}{2}h(\phi)$

Step 1: enthalpy of each phase

Enthalpy per unit volume is stored heat: sensible heat c_pT plus latent heat. At the same temperature the liquid holds an extra L:

$$H_{\text{solid}} = c_pT, \qquad H_{\text{liquid}} = c_pT + L.$$

Step 2: interpolate with the phase field

Inside a diffuse interface a point is partly solid. Let $h(\phi)$ run from -1 (liquid) to $+1$ (solid). Then the liquid fraction is

$$\text{liquid fraction} = \frac{1-h(\phi)}{2} \qquad (\text{liquid: } 1,\ \text{solid: } 0),$$

$$H = c_pT + L\,\frac{1-h(\phi)}{2}.$$

Step 3: non-dimensionalise

With $u = c_p(T - T_m)/L$, i.e. $c_pT = c_pT_m + Lu$, divide by L:

$$\frac{H}{L} = \underbrace{\frac{c_pT_m}{L} + \frac{1}{2}}_{\text{constant}} + u - \frac{h(\phi)}{2}.$$

Only changes of enthalpy matter, so drop the constant and define

$$\boxed{U = u - \tfrac{1}{2}h(\phi)}$$

In the solid $U = u - \frac{1}{2}$; in the liquid $U = u + \frac{1}{2}$: the liquid holds exactly one more unit (L, in scaled form). The factor $\frac{1}{2}$ appears because h spans 2 units while the latent heat is 1 unit. With Bhagat's choice $h = \phi$ this is $U = u - \frac{1}{2}\phi$.

Physical check. Freeze a point so fast that heat cannot leave (U fixed): h rises by 2, so u rises by 1, i.e. T rises by L/c_p. All released latent heat has warmed the material, as it should.

4 Why U is used to write the temperature equation

1. **U is conserved; u is not.** Total enthalpy changes only by heat flowing across the boundary. Temperature, by contrast, jumps up wherever latent heat is released, without any heat arriving. A conservation law is naturally written for a conserved quantity:
$$\frac{\partial U}{\partial t} = -\nabla \cdot \mathbf{J}.$$
2. **Latent heat is handled automatically.** Written for U, the equation has no source term. The latent-heat source appears only when it is rewritten in u (Section 5). Using U guarantees that exactly one unit of latent heat is released per unit of solid formed.
3. **It is the correct variable for the variational structure.** A conserved field evolves by a *conserved* gradient flow $\partial_t U = \nabla \cdot (M \nabla \, \delta \Pi_L / \delta U)$, while the non-conserved ϕ relaxes directly. And the physically correct variation of ϕ is one at *fixed* U: when a bit of liquid freezes, its enthalpy cannot change instantly, but its temperature does. This is exactly what brings h' into the coupling term (Section 6).

5 Derivation of Eq. 4a: $\partial_t U = \nabla \cdot \left(D \nabla \, \delta \Pi_L / \delta u \right)$

Route 1: energy conservation

Energy conservation with Fourier's law $\mathbf{q} = -k \nabla T$ gives $\partial_t H = \nabla \cdot (k \nabla T)$. Divide by L and use $c_p T = c_p T_m + L u$ (k, c_p constant, $D = k/c_p$):
$$\frac{\partial U}{\partial t} = D \nabla^2 u. \tag{1}$$
Heat flows only where temperature varies, and only enthalpy is conserved.

Route 2: as a conserved gradient flow of Π_L

Take the variational derivative of Π_L with respect to U at fixed ϕ. Only the ξu^2 term depends on U, and $\partial u / \partial U|_\phi = 1$:
$$\left. \frac{\delta \Pi_L}{\delta U} \right|_\phi = 2 \xi u \left. \frac{\partial u}{\partial U} \right|_\phi = 2 \xi u.$$
So temperature is the "chemical potential" of enthalpy: $u = \frac{1}{2\xi} \, \delta \Pi_L / \delta U$. Substituting into (1),
$$\frac{\partial U}{\partial t} = \nabla \cdot \left(\frac{D}{2\xi} \nabla \frac{\delta \Pi_L}{\delta U} \right).$$
Bhagat et al. write $\delta \Pi_L / \delta u$; at fixed ϕ this equals $\delta \Pi_L / \delta U$ because $\partial u / \partial U = 1$. The constant $1/(2\xi)$ is absorbed into the normalisation (Karma & Rappel write $\frac{b\lambda}{2} u^2$ in F, giving $\partial_t U = \frac{D}{b\lambda} \nabla^2 \, \delta F / \delta U$, which is exactly (1)). This is **Eq. 4a**. It is pure energy conservation written in the language of Π_L; nothing else is assumed.

Why this form is "downhill". With no flux through the boundary, integrating by parts,
$$\int_\Omega \frac{\delta \Pi_L}{\delta U} \frac{\partial U}{\partial t} \, dV = -\frac{D}{2\xi} \int_\Omega \left| \nabla \frac{\delta \Pi_L}{\delta U} \right|^2 dV \leq 0,$$
so heat diffusion alone can only lower Π_L.

From Eq. 4a to Eq. 5a

Substitute $U = u - \frac{1}{2}h(\phi)$ into (1):

$$\frac{\partial u}{\partial t} = D\nabla^2 u + \frac{1}{2}h'(\phi)\frac{\partial \phi}{\partial t} \xrightarrow{h=\phi} \boxed{\frac{\partial u}{\partial t} = D\nabla^2 u + \frac{1}{2}\frac{\partial \phi}{\partial t}} \quad \text{(Eq. 5a)}$$

The last term is the latent heat released where ϕ is changing, i.e. at the moving interface—the diffuse-interface version of the Stefan condition (Eq. 1b).

6 Approach 1: one functional for everything

Suppose we insist on a single free energy functional for consistency. It must be the Lyapunov functional Π_L, because it is the one that always decreases as the coupled ϕ–U system evolves:

$$\tau\frac{\partial \phi}{\partial t} = -\frac{\delta \Pi_L}{\delta \phi}\bigg|_U, \qquad \frac{\partial U}{\partial t} = \nabla \cdot \left(M\nabla \frac{\delta \Pi_L}{\delta U}\bigg|_\phi\right).$$

The coupling term

Vary ϕ at fixed U. Since $u = U + \frac{1}{2}h(\phi)$, the term ξu^2 changes too (chain rule):

$$\frac{\partial}{\partial \phi}\left[\xi u^2\right]_U = 2\xi u \left.\frac{\partial u}{\partial \phi}\right|_U = 2\xi u \cdot \tfrac{1}{2}h'(\phi) = \xi\, u\, h'(\phi).$$

The phase-field equation becomes

$$\tau\frac{\partial \phi}{\partial t} = -g'(\phi) - \underbrace{\xi\, u\, h'(\phi)}_{\text{coupling term}} + \nabla \cdot (\lambda^2 \nabla \phi) + \dots$$

For $u < 0$ the coupling term pushes ϕ up, towards solid, as it should.

The problem with $h = \phi$

With Bhagat's $h = \phi$, $h' = 1$ and the coupling is simply ξu. Then

$$g'(\phi) + \xi u = -\phi + \phi^3 + \xi u \neq 0 \quad \text{at } \phi = \pm 1 \text{ whenever } u \neq 0,$$

so the valleys shift away from ± 1 and the bulk solid and liquid values drift with temperature. This is undesirable (older models tolerated it; Karma & Rappel call fixed minima an "advantage").

The fix within Approach 1: a different h

Keeping the minima at ± 1 with one functional requires $h'(\phi) \propto (1-\phi^2)^2$. Karma & Rappel's variational form (VF) uses

$$h_{VF}(\phi) = \tfrac{15}{8}\left(\phi - \tfrac{2}{3}\phi^3 + \tfrac{1}{5}\phi^5\right), \qquad h_{VF}(\pm 1) = \pm 1,$$

so exactly one unit of latent heat is still released. Approach 1 is therefore possible—but only with a 5th-order latent-heat function, and the heat equation then reads $\partial_t u = D\nabla^2 u + \frac{1}{2}h'_{VF}(\phi)\,\partial_t \phi$.

What Approach 1 buys: guaranteed decrease

With no boundary flux,

$$\frac{d\Pi_L}{dt} = \int_\Omega \Big[\frac{\delta\Pi_L}{\delta\phi}\frac{\partial\phi}{\partial t} + \frac{\delta\Pi_L}{\delta U}\frac{\partial U}{\partial t}\Big]dV = -\int_\Omega \tau\Big(\frac{\partial\phi}{\partial t}\Big)^2 dV - \int_\Omega M\Big|\nabla\frac{\delta\Pi_L}{\delta U}\Big|^2 dV \le 0.$$

Both terms are non-positive, so Π_L always decreases: the model is thermodynamically consistent.

7 Approach 2: isothermal functional for ϕ, conservation for u

Use a separate *isothermal free energy functional* Π for the phase field, with u treated as a fixed parameter:

$$\tau\frac{\partial\phi}{\partial t} = -\frac{\delta\Pi}{\delta\phi} = -g'(\phi) - \underbrace{\xi\, u\,(1-\phi^2)^2}_{\text{coupling term}} + \nabla\cdot(\lambda^2\nabla\phi) + \dots$$

The coupling now comes from $p'(\phi) = (1-\phi^2)^2$, which vanishes at $\phi = \pm 1$: **the equilibrium values stay at ± 1 for every temperature**, even though $h = \phi$. The temperature equation is energy conservation (Section 5) with $h = \phi$. The tilt p and the latent heat h are now chosen independently.

Note that Π is not literally Π_L held at constant temperature: Π_L contains ξu^2, while Π contains $\xi u\, p(\phi)$, linear in u. It is a separate functional in which u is a parameter.

The cost: no guaranteed decrease

$$\frac{d\Pi}{dt} = \int_\Omega \frac{\delta\Pi}{\delta\phi}\frac{\partial\phi}{\partial t}dV + \int_\Omega \xi\, p(\phi)\frac{\partial u}{\partial t}dV = -\int_\Omega \tau\Big(\frac{\partial\phi}{\partial t}\Big)^2 dV + \underbrace{\int_\Omega \xi\, p(\phi)\frac{\partial u}{\partial t}dV}_{\text{either sign}}.$$

The first term is always ≤ 0: ϕ always moves downhill. The second term comes from temperature changing; it has no definite sign. Picture a ball rolling downhill while someone keeps tilting the ground.

"Only variational isothermally in the trivial case"

Karma & Rappel: in this formulation "the phase-field equations are not derived from a single Lyapounov functional. They are only variational isothermally in the trivial case where u is taken to be constant." If u is held constant, $\partial_t u = 0$, the second term vanishes and Π always decreases. But constant u means no heat flow and no latent heat—exactly the physics of solidification is switched off. Hence "trivial".

How to deal with it

Nothing needs to be done. A monotonically decreasing functional is a strong theoretical property (useful for proofs and energy-stable time schemes), but it is not required for correct physics. Approach 2 still conserves energy exactly, relaxes ϕ downhill at the local temperature, and reproduces the correct Gibbs–Thomson and Stefan conditions in the thin-interface limit—the property that is actually checked.

8 Comparing the two approaches

	Approach 1 (VF, one functional)	**Approach 2** (IVF, used by Bhagat)
ϕ equation	$-\delta\Pi_L/\delta\phi\|_U$	$-\delta\Pi/\delta\phi$
Temperature equation	$\partial_t U = \nabla\cdot(M\nabla\,\delta\Pi_L/\delta U)$	energy conservation (same structure)
Coupling term	$\xi u\, h'(\phi)$, tied to h	$\xi u\, p'(\phi) = \xi u(1-\phi^2)^2$
Latent-heat function	h_{VF} (5th order)	$h = \phi$
Heat source	$\frac{1}{2}h'_{VF}\,\partial_t\phi$	$\frac{1}{2}\partial_t\phi$
Minima at ±1	yes (with h_{VF})	yes
Functional always decreases	yes	not guaranteed
Grid convergence	slower	faster ($\Delta x/W_0 = 0.4$ suffices)
High-velocity behaviour	can give a spurious steady-state branch	no spurious branch reported
Thin-interface limit	Stefan problem	same Stefan problem

Karma & Rappel showed both reduce to the same sharp-interface Stefan problem as the interface becomes thin. They call the VF "thermodynamically more consistent" but argue that the choice should rest purely on computational efficiency and accuracy, since the model's quantitative content lies only in d_0 and β.

Why Approach 2 is computationally cheaper

1. **Per time step, the cost is about the same.** Both solve one ϕ equation and one heat equation.
2. **The saving is in grid resolution.** Karma & Rappel refined $\Delta x/W_0$ in steps of 0.1 until the tip velocity changed by less than 2%. The IVF ($h = \phi$) converged at a coarser grid than the VF (their Fig. 5), and they used $\Delta x/W_0 = 0.4$.
3. **A coarser grid saves a lot.** Doubling Δx in 3D gives $2^3 = 8$ times fewer nodes; with explicit time stepping $\Delta t \propto \Delta x^2$, so 4 times fewer steps: roughly 32 times less work.
4. **Fewer failure modes.** At high velocity the VF can produce an unphysical spurious branch of steady-state solutions.

Karma & Rappel report this as a numerical finding and do not explain its cause. A plausible reading (not theirs): $h'_{VF} \propto (1-\phi^2)^2$ concentrates the latent-heat release sharply in the middle of the interface, and a sharper source needs a finer grid; $h = \phi$ spreads the release more evenly.

9 How Bhagat et al. fit in

Bhagat et al. use Approach 2:

$$\frac{\partial U}{\partial t} = \nabla \cdot \left(D \nabla \frac{\delta \Pi_L}{\delta u} \right) \text{(4a)}, \qquad \tau(\mathbf{n}) \frac{\partial \phi}{\partial t} = -\frac{\delta \Pi}{\delta \phi} \text{(4b)}.$$

- Eq. 4a is written with Π_L, but it is only energy conservation in variational notation (Section 5); it would be identical in either approach for the same h.
- Eq. 4b uses the isothermal Π, not Π_L. Its coupling $\xi u(1-\phi^2)^2$ (from Eq. 3b) differs from the ξu that Π_L would give with $h = \phi$. That mismatch is exactly the IVF.
- Hence the coupled system is not derived from a single functional; the paper mentions Π_L and refers to [13, 41, 42] for details.

10 What the three references contribute

[42] Karma & Rappel (1998), Phys. Rev. E 57, 4323.
The source of the whole argument (Sec. II): defines the VF and the IVF, shows both give the same free-boundary problem, reports the IVF's faster grid convergence and the VF's spurious branch. Also gives the thin-interface relations $d_0 = a_1 W/\lambda$ and $\beta = a_1(\tau/\lambda W - a_2 W/D)$, the origin of $a_1 = 0.8839$, $a_2 = 0.6267$ in Bhagat's Table 3.

[13] Plapp & Karma (2000), J. Comput. Phys. 165, 592.
Uses the IVF as is: a Ginzburg–Landau free energy for ψ with the same f as Bhagat Eq. 3b, and the heat equation with a latent-heat source. Never says "Lyapunov" or "isothermal"; its contribution is a hybrid diffusion–Monte Carlo method.

[41] Boettinger, Warren, Beckermann & Karma (2002), Annu. Rev. Mater. Res. 32, 163.
The general recipe. The fully consistent route starts from an entropy functional $S(e, c, \phi)$ whose entropy always increases. The common shortcut uses an isothermal free energy $F(\phi, c, T)$ for ϕ (Allen–Cahn) and solute (Cahn–Hilliard) and appends a heat-flow equation $C_p \partial_t T + L \partial_t \phi = \nabla \cdot (k \nabla T)$, with essentially equivalent results.

11 Quick recall

1. g makes two phases; p lets temperature pick one; h says how much latent heat each point holds.
2. $U = u - \frac{1}{2}h(\phi)$ is the total heat content; it is conserved, so the heat equation is written for U.
3. Eq. 4a is energy conservation written as a conserved gradient flow of Π_L ($\delta \Pi_L / \delta U = 2\xi u$).
4. One functional (Π_L) gives coupling $\xi u h'(\phi)$: with $h = \phi$ the minima drift; with h_{VF} they don't, but convergence is slower.
5. The isothermal Π gives coupling $\xi u(1-\phi^2)^2$ with $h = \phi$: minima fixed, faster convergence, no guaranteed decrease (variational only in the trivial isothermal case).
6. Both give the same Stefan problem as the interface thins; Bhagat et al. use the cheaper Approach 2.

Corrections made during the discussion

- Eq. 4a *is* written as a variational derivative of Π_L; only the coupled system is not derived from one functional.
- Karma & Rappel did not break the variational structure to obtain the thin-interface limit (which works either way); the reasons are independent p and h and better numerical convergence.
- Approach 1 is not impossible: it works with h_{VF} instead of $h = \phi$.
- The two approaches share the structure of the heat equation, but its source term depends on h.

Sources

- Bhagat et al. (2023), *Engineering with Computers* 39:2345–2363 — Sect. 2.2, Eqs. 3–5, Table 3.
- Karma & Rappel (1998), *Phys. Rev. E* 57, 4323 — Sec. II, Sec. V, Fig. 5. doi:10.1103/PhysRevE.57.4323
- Plapp & Karma (2000), *J. Comput. Phys.* 165, 592 — Eqs. 9–11, Appendix. arXiv:cond-mat/0002015
- Boettinger et al. (2002), *Annu. Rev. Mater. Res.* 32, 163 — Eqs. 1–8. doi:10.1146/annurev.matsci.32.101901.155803